Lecture 8. Exceptions, Processes, Threads

Control Flow

- **Processors do only one thing:**
  - From startup to shutdown, a CPU simply reads and executes (interprets) a sequence of instructions, one at a time
  - This sequence is the CPU's *control flow* (or *flow of control*)

Physical control flow

Time

<startup>

inst_1

inst_2

inst_3

…

inst_n

<shutdown>

Altering the Control Flow

- **Up to now: two mechanisms for changing control flow:**
  - Jumps and branches
  - Call and return

  Both react to changes in ***program state***

- **Insufficient for a useful system:**
  Difficult to react to changes in *system state*
  - data arrives from a disk or a network adapter
  - instruction divides by zero
  - user hits Ctrl-C or Ctrl-Alt-Del at the keyboard
  - System timer expires

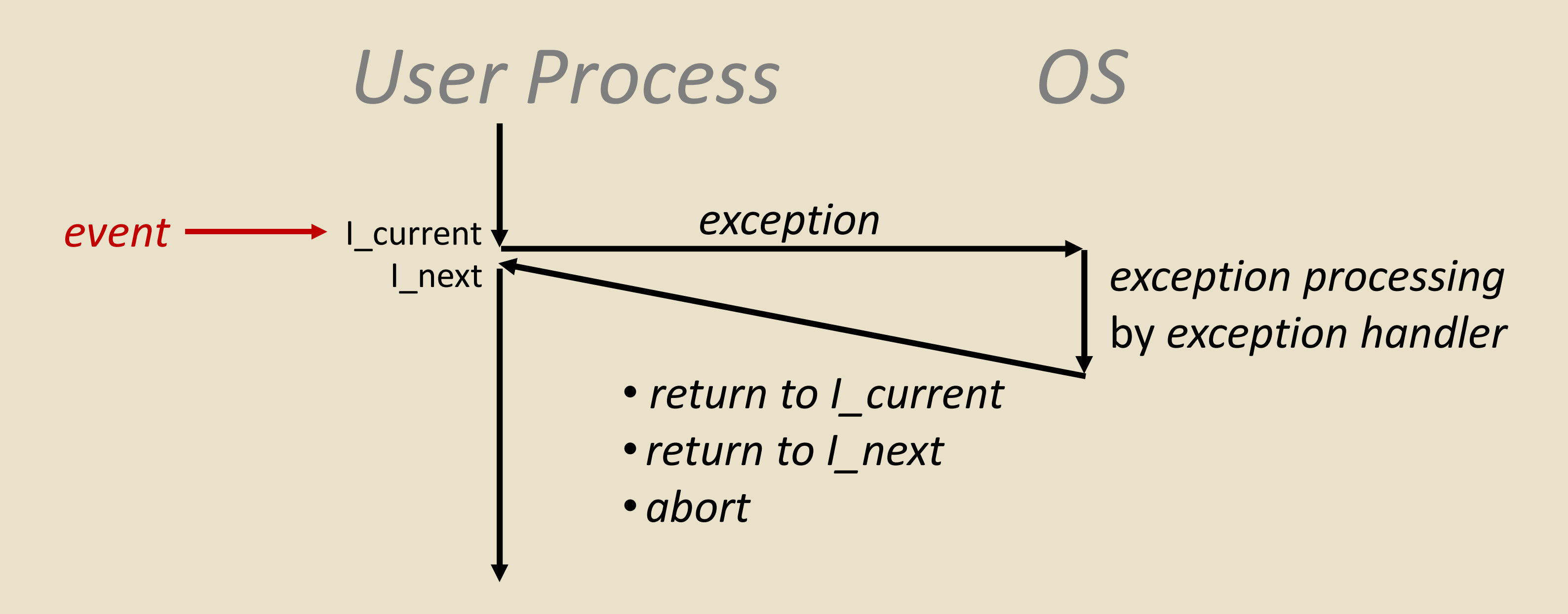
- **System needs mechanisms for "exceptional control flow"**

Exceptional Control Flow

- **Exists at all levels of a computer system**
- **Low level mechanisms**
  - Exceptions
    - change in control flow in response to a system event (i.e., change in system state)
  - Combination of hardware and OS software
- **Higher level mechanisms**
  - Process context switch
  - Signals
  - Nonlocal jumps: setjmp()/longjmp()
  - Implemented by either:
    - OS software (context switch and signals)
    - C language runtime library (nonlocal jumps)

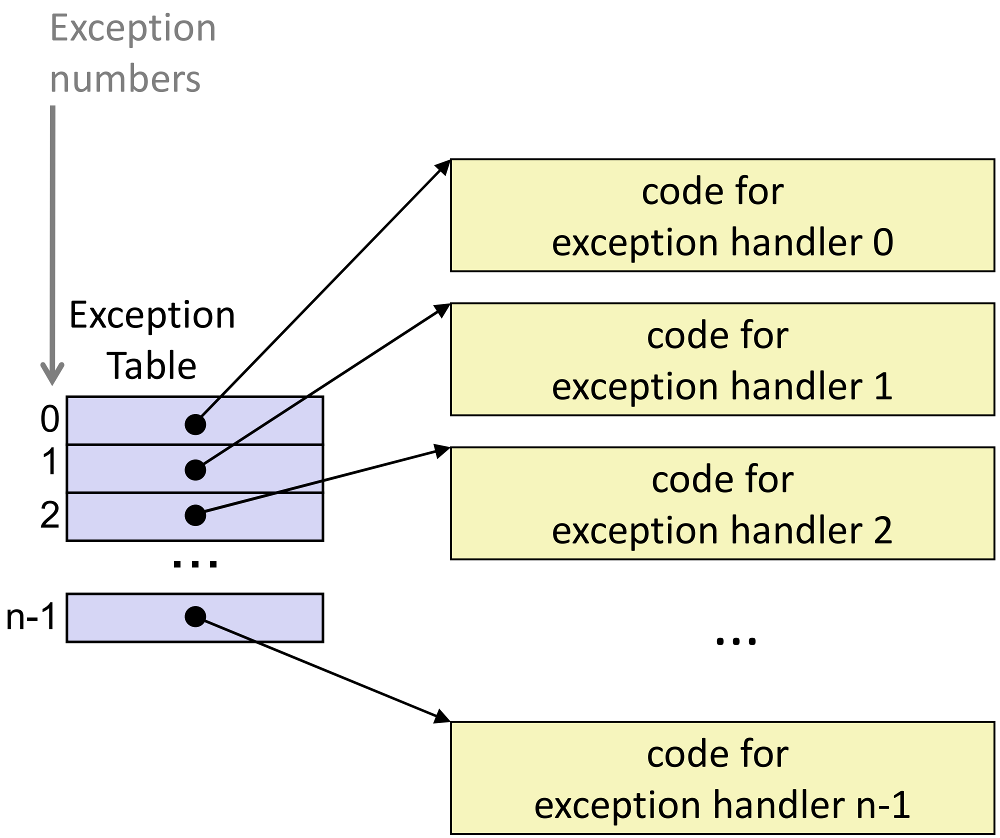
Exceptions

- **An *exception* is a transfer of control to the OS in response to some *event* (i.e., change in processor state)**



- **Examples:**
  div by 0, arithmetic overflow, page fault, I/O request completes, Ctrl-C

Exception Tables



- **Each type of event has a unique exception number k**
- **k = index into exception table (a.k.a. interrupt vector)**
- **Handler k is called each time exception k occurs**

Asynchronous Exceptions (Interrupts)

- **Caused by events external to the processor**
  - Indicated by setting the processor's interrupt pin
  - Handler returns to "next" instruction
- **Examples:**
  - I/O interrupts
    - hitting Ctrl-C at the keyboard
    - arrival of a packet from a network
    - arrival of data from a disk
  - Hard reset interrupt
    - hitting the reset button
  - Soft reset interrupt
    - hitting Ctrl-Alt-Delete on a PC

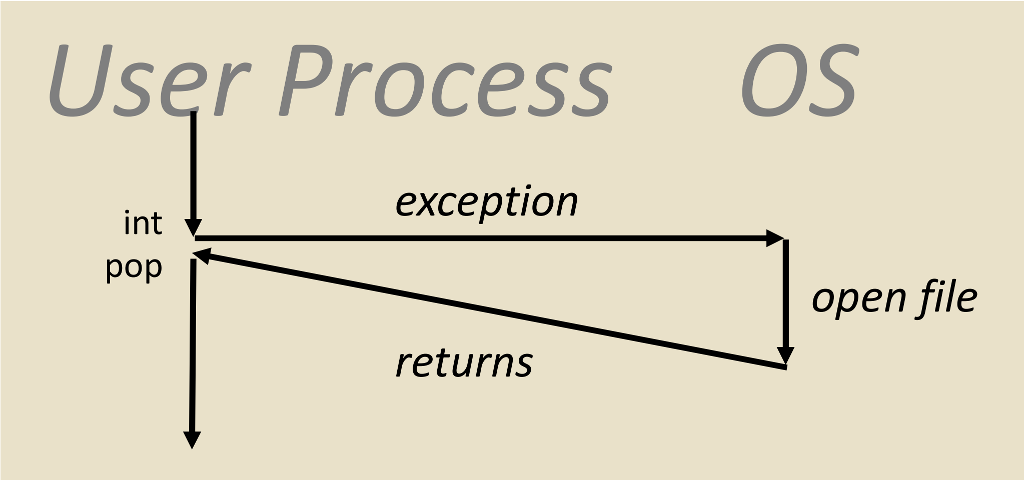
Synchronous Exceptions

- **Caused by events that occur as a result of executing an instruction:**
  - ***Traps***
    - Intentional
    - Examples: ***system calls***, breakpoint traps, special instructions
    - Returns control to "next" instruction
  - ***Faults***
    - Unintentional but possibly recoverable
    - Examples: page faults (recoverable), protection faults (unrecoverable), floating point exceptions
    - Either re-executes faulting ("current") instruction or aborts
  - ***Aborts***
    - unintentional and unrecoverable
    - Examples: parity error, machine check
    - Aborts current program

Trap Example: Opening File

- User calls: **`open(filename, options)`**
- Function **`open`** executes system call instruction **`int`**

```
0804d070 <__libc_open>:
 . . .
 804d082:       cd 80                  int     $0x80
 804d084:       5b                     pop     %ebx
 . . .
```



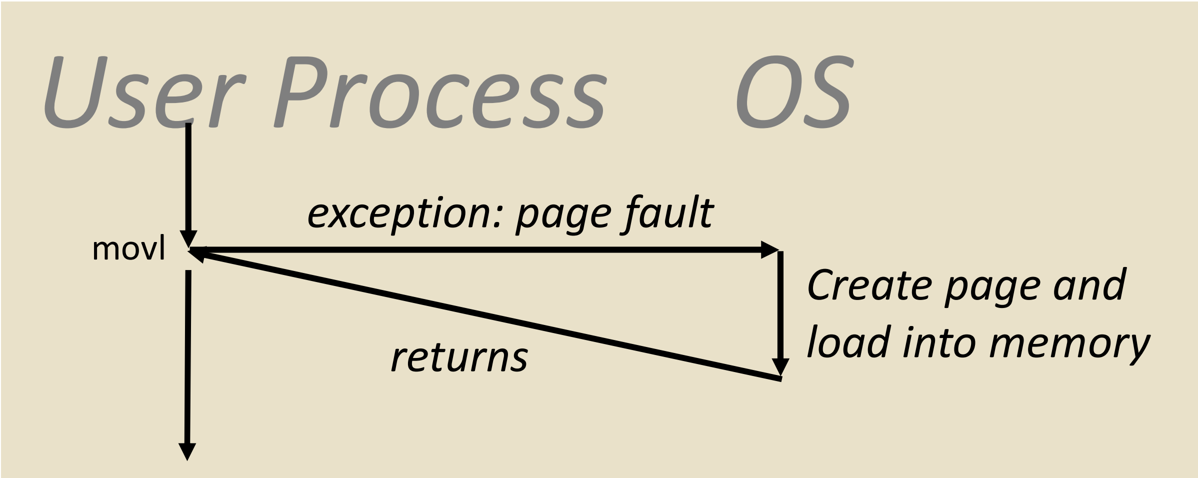
- OS must find or create file, get it ready for reading or writing
- Returns integer file descriptor

Fault Example: Page Fault

- User writes to memory location
- That portion (page) of user's memory is currently on disk

```
int a[1000];
main ()
{
    a[500] = 13;
}
```

```
80483b7:        c7 05 10 9d 04 08 0d    movl    $0xd,0x8049d10
```

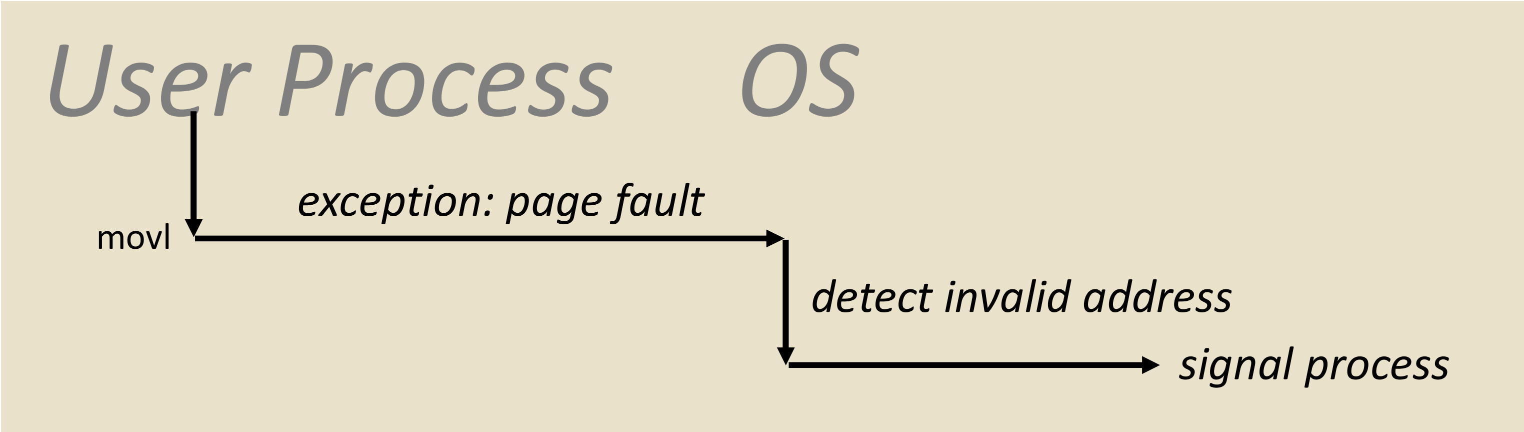


- Page handler must load page into physical memory
- Returns to faulting instruction
- Successful on second try

Fault Example: Invalid Memory Reference

```
int a[1000];
main ()
{
    a[5000] = 13;
}
```

```
80483b7:        c7 05 60 e3 04 08 0d    movl    $0xd,0x804e360
```



- Page handler detects invalid address
- Sends **`SIGSEGV`** signal to user process
- User process exits with “segmentation fault”

Exception Table IA32 (Excerpt)

Exception Number	Description	Exception Class
0	Divide error	Fault
13	General protection fault	Fault
14	Page fault	Fault
18	Machine check	Abort
32-127	OS-defined	Interrupt or trap
128 (0x80) 33 (0x21)	System call (Windows)	Trap Trap
129-255	OS-defined	Interrupt or trap

Check Table 6-1:
http://download.intel.com/design/processor/manuals/253665.pdf